In p-block, the elements present in lower left part are metals while the elements in the upper right part are non metals. Elements of group 13 have metallic character except the first element boron which is a metalloid, having properties intermediate between the metal and nonmetals. The atomic radius of boron is very small and it has relatively high nuclear charge and these properties are responsible for its nonmetallic character. In the subsequent groups the non-metallic character increases. In group 14 elements, carbon is a nonmetal while silicon and germanium are metalloids. In group 15, nitrogen and phosphorus are non metals and arsenic & antimony are metalloids. In group 16, oxygen, sulphur and selenium are non metals and tellurium is a metalloid. All the elements of group 17 and 18 are non metals.

2.1.3 Ionisation Enthalpy:

We have already learnt that as we move down a group, generally there is a steady decrease in ionisation enthalpy of elements due to increase in their atomic radius. In p-block elements, there are some minor deviations to this general trend. In group 13, from boron to aluminium the ionisation enthalpy decreases as expected. But from aluminium to thallium there is only a marginal difference. This is due to the presence of inner d and f-electrons which has poor shielding effect compared to s and p-electrons. As a result, the effective nuclear charge on the valance electrons increases. A similar trend is also observed in group 14. The remaining groups (15 to 18) follow the general trend. In these groups, the ionisation enthalpy decreases, as we move down the group. Here, poor shielding effect of d- and f-electrons are overcome by the increased shielding effect of the additional p-electrons. The ionisation enthalpy of elements in successive groups is higher than the corresponding elements of the previous group as expected.

2.1.4 Electronegativity

As we move down the 13^{th} group, the electronegativity first decreases from boron to aluminium and then marginally increases for Gallium, thereafter there is no appreciable change. Similar trend is also observed in 14^{th} group as well. In other groups, as we move down the group, the electro negativity decreases. This observed trend can be correlated with their atomic radius.

2.1.5 Anomalous properties of the first elements:

In p-block elements, the first member of each group differs from the other elements of the corresponding group. The following factors are responsible for this anomalous behaviour.

1. Small size of the first member
2. High ionisation enthalpy and high electronegativity
3. Absence of d orbitals in their valance shell

The first member of the group 13, boron is a metalloid while others are reactive metals. Moreover, boron shows diagonal relationship with silicon of group 14. The oxides of boron and silicon are similar in their acidic nature. Both boron and silicon form covalent hydrides that can be easily hydrolysed. Similarly, except boron trifluoride, halides of both elements are readily hydrolysed.

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